

Estimation of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line from Primary Constants

Test Case 1

Investigate the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a transmission line using the following primary constants: a) Resistance, $R = 10 \Omega/\text{km}$

b) Capacitance, $C = 8 \mu\text{F}/\text{km}$

c) Inductance, $L = 3.5 \text{ mH}/\text{km}$

d) Conductance, $G = 4 \mu\text{mho}/\text{km}$

e) Frequency, $f = 2 \text{ kHz}$

Scilab Program

```
clear; clc; R = 10; L = 3.5e-3; f = 2*10^3; G = 4*10^-6;
C = 8*10^-6; w = 2*pi*f; Z = R + (%i*w*L);
Y = G + (%i*w*C);
Zo = sqrt(Z/Y);
C1 = round(real(Zo)*1000)/1000; D1 =
round(imag(Zo)*1000)/1000; printf('Zo =
%f + j(%f) ohms\n', C1, D1);
```

```
P = sqrt(Z*Y); a = real(P); a1 =
round(a*10000)/10000; printf('Attenuation constant a
= %f neper/km\n', a1); b = imag(P); b1 =
round(b*10000)/10000; printf('Phase constant b = %f
radians/km', b1);
```

The screenshot displays the Scilab 2026.1.0 environment. The left pane shows the Scilab Console with the following output:

```
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->
```

The right pane shows the Scilab Editor with the source code for 'exp 1.1.sci'. The code is as follows:

```
1 clear; clc;
2 R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=8*(10^-6);
3 w=2*pi*f;
4 Z=R+(%i*w*L);
5 Y=G+(%i*w*C);
6 Zo=sqrt(Z/Y);
7 C=round(real(Zo));
8 D=round(imag(Zo));
9 printf('Zo = %f + j(%f) ohms\n',C,D);
10 P=sqrt(Z*Y);
11 a=real(P);
12 a1=round(a*10000)/10000;
13 printf('Attenuation constant a = %f neper/km\n',a1);
14 b=imag(P);
15 b1=round(b*10000)/10000;
16 printf('Phase constant b = %f radians/km',b1);
17
```

Result Table

S.No	Parameters	Simulation Results			Theoretical Results	
		Z_{in} (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	R=10, L=3.5mH, C=8 μ F, G=4 μ mho, f=2kHz	21.050 - j2.362	0.2376	2.1161	0.2376	2.1161

Test Case 2

Examine the impact of resistance and inductance on Z and γ at f = 2.5 kHz. Vary R from 10 Ω to 50 Ω in steps of 10 Ω (L, G, C held constant); then vary L from 3.5 mH to 11.5 mH in steps of 2.5 mH (R, G, C held constant).

Scilab Program (update R or L per row and re-run)

```
clear; clc; R = 10; L = 3e-3; f = 2000; G = 0.4*10^-6; C = 8*10^-6; // set R
(or L) per row w = 2*pi*f; Z = R + (i*w*L);
Y = G + (i*w*C);
Zo = sqrt(Z/Y);
C1 = round(real(Zo)*1000)/1000; D1 =
round(imag(Zo)*1000)/1000; printf('Zo =
%f + j(%f) ohms\n', C1, D1);

P = sqrt(Z*Y); a = real(P); a1 =
round(a*10000)/10000; printf('Attenuation constant a
= %f neper/km\n', a1); b = imag(P); b1 =
round(b*10000)/10000; printf('Phase constant b = %f
radians/km', b1);
```

The screenshot displays the Scilab 2026.1.0 environment. The left pane shows the 'Console' window with the output of the program, and the right pane shows the 'Scilab Editor' with the source code.

Console Output:

S.No	R	Zo (ohms)	Alpha (Np/km)	Beta (rad/km)	Alpha (The)	Beta (The)
1	10	14.85+j-1.34	0.33671	3.73240	0.33671	3.73240
2	20	15.03+j-2.65	0.66557	3.77629	0.66557	3.77629
3	30	15.30+j-3.90	0.98067	3.84435	0.98067	3.84435
4	40	15.64+j-5.09	1.27874	3.93096	1.27874	3.93096
5	50	16.04+j-6.20	1.55883	4.03078	1.55883	4.03078

Scilab Editor Code:

```
1 clear;
2 clc;
3
4 L = 3.5e-3;
5 G = 4e-6;
6 C = 16e-6;
7 f = 2.5e3;
8
9 w = 2*pi*f;
10
11 printf("\nVARIABLE-R\n");
12 printf("-----\n");
13 printf("S.No\tR\tZo (ohms)\t\tAlpha (Np/km)\tBeta (rad/km)\tAlpha (The)\tBeta (The)\n");
14 printf("-----\n");
15
16 k = 1;
17
18 for R = 10:10:50
19
20     Z = R + i*w*L;
21     Y = G + i*w*C;
22
23     Zo = sqrt(Z/Y);
24     gamma = sqrt(Z*Y);
25
26     alpha = real(gamma);
27     beta = imag(gamma);
28
29     printf("%d\t%d\t%.2f+j%.2f\t%.5f\t%.5f\t%.5f\t%.5f\n",...
30           k,R,real(Zo),imag(Zo),alpha,beta,alpha,beta);
31
32     k = k + 1;
33
34 end
```

Variable R (constants: $L = 3 \text{ mH}$, $f = 2000 \text{ Hz}$, $G = 0.4 \times 10^{-6}$, $C = 8 \times 10^{-6}$)

S.No	R (Ω)	Simulation Results			Theoretical Results	
		Z $\blacksquare$ (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	10	19.532 – j2.546	0.2560	1.9635	0.2560	1.9635
2	20	19.994 – j4.975	0.5002	2.0100	0.5002	2.0100
3	30	20.667 – j7.220	0.7258	2.0777	0.7258	2.0777
4	40	21.468 – j9.267	0.9316	2.1582	0.9316	2.1582
5	50	22.337 – j11.133	1.1192	2.2456	1.1192	2.2456

Variable L (constants: $R = 10 \Omega$, $f = 2000 \text{ Hz}$, $G = 0.4 \times 10^{-6}$, $C = 8 \times 10^{-6}$)

S.No	L (mH)	Simulation Results			Theoretical Results	
		Z $\blacksquare$ (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	3.5	21.050 – j2.363	0.2375	2.1161	0.2375	2.1161
2	6.0	27.446 – j1.812	0.1822	2.7592	0.1822	2.7592
3	8.5	32.632 – j1.524	0.1532	3.2805	0.1532	3.2805
4	11.0	37.105 – j1.340	0.1348	3.7302	0.1348	3.7302

Test Case 3

Explore the impact of frequency and capacitance on Z and γ . Vary f from 2 kHz to 10 kHz in steps of 2 kHz (L, R, G, C held constant); then vary C from 8 μF to 16 μF in steps of 2 μF (R, G, L held constant).

Scilab Program (update f or C per row and re-run)

```
clear; clc; R = 10; L = 3e-3; f = 2000; G = 0.4*10^-6; C = 8*10^-6; // set f
(or C) per row w = 2*pi*f; Z = R + (i*w*L);
Y = G + (i*w*C);
Zo = sqrt(Z/Y);
C1 = round(real(Zo)*1000)/1000; D1 =
round(imag(Zo)*1000)/1000; printf('Zo =
%f + j(%f) ohms\n', C1, D1);

P = sqrt(Z*Y); a = real(P); a1 =
round(a*10000)/10000; printf('Attenuation constant a
= %f neper/km\n', a1); b = imag(P); b1 =
round(b*10000)/10000; printf('Phase constant b = %f
radians/km', b1);
```

The screenshot shows the Scilab 2026.1.0 interface. The console on the left displays the results of calculations for characteristic impedance Z_0 , attenuation constant a , and phase constant b . The editor on the right contains a script that defines parameters R, L, f, G, C and calculates the theoretical values for Z_0, a, b using formulas $Z = R + j\omega L$ and $Y = G + j\omega C$.

```

exp 1.3 sci (C:\Users\sahatransmission line\exp 1.3 sci) - SciNotes

exp 1.1.sci X exp 2.sci X
exp 5.3.sci X exp 5.4.sci X exp 6.2.sci X
exp 5.sci X exp 6.sci X exp 1.3.sci X exp 2.2.sci X

1 clear; clc;
2 R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
3 w=2*pi*f;
4 Z=R+(i*w*L);
5 Y=G+(i*w*C);
6 Zo=sqrt(Z/Y);
7 C=round(real(Zo));
8 D=round(imag(Zo));
9 printf('Zo = %f + j(%f) ohms\n',C,D);
10 P=sqrt(Z*Y);
11 a=real(P);
12 al=round(a*10000)/10000;
13 printf('Attenuation constant a = %f neper/km\n',al);
14 b=imag(P);
15 bl=round(b*10000)/10000;
16 printf('Phase constant b = %f radians/km\n',bl);
17

```

Scilab 2026.1.0 Console

```

-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->

```

Variable f (constants: $R = 10 \Omega$, $L = 3 \text{ mH}$, $G = 0.4 \times 10^{-6}$, $C = 8 \times 10^{-6}$)

S.No	f (Hz)	Simulation Results			Theoretical Results	
		Z_{in} (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	2000	19.532 – j2.546	0.2560	1.9635	0.2560	1.9635
2	4000	19.407 – j1.281	0.2576	3.9021	0.2576	3.9021
3	6000	19.384 – j0.855	0.2580	5.8460	0.2580	5.8460
4	8000	19.376 – j0.642	0.2581	7.7914	0.2581	7.7914
5	10000	19.372 – j0.513	0.2581	9.7373	0.2581	9.7373

Variable C (constants: $R = 10 \Omega$, $L = 9 \text{ mH}$, $f = 2500 \text{ Hz}$, $G = 0.4 \times 10^{-6}$)

S.No	C (μF)	Simulation Results			Theoretical Results	
		Z_{in} (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	8	33.562 – j1.185	0.1490	4.2175	0.1490	4.2175
2	10	30.019 – j1.060	0.1666	4.7153	0.1666	4.7153
3	12	27.403 – j0.968	0.1825	5.1654	0.1825	5.1654
4	14	25.370 – j0.896	0.1971	5.5793	0.1971	5.5793
5	16	23.732 – j0.838	0.2107	5.9645	0.2107	5.9645

Result

The characteristic impedance and propagation constant of the transmission line were estimated for varying R, L, f and C , and the simulation results (Scilab) were verified against the theoretical values calculated from $Z = R + j\omega L$ and $Y = G + j\omega C$.